

1 Background Theory

1.1 Differential amplifier

The differential amplifier single input balanced output is shown in Figure ??.

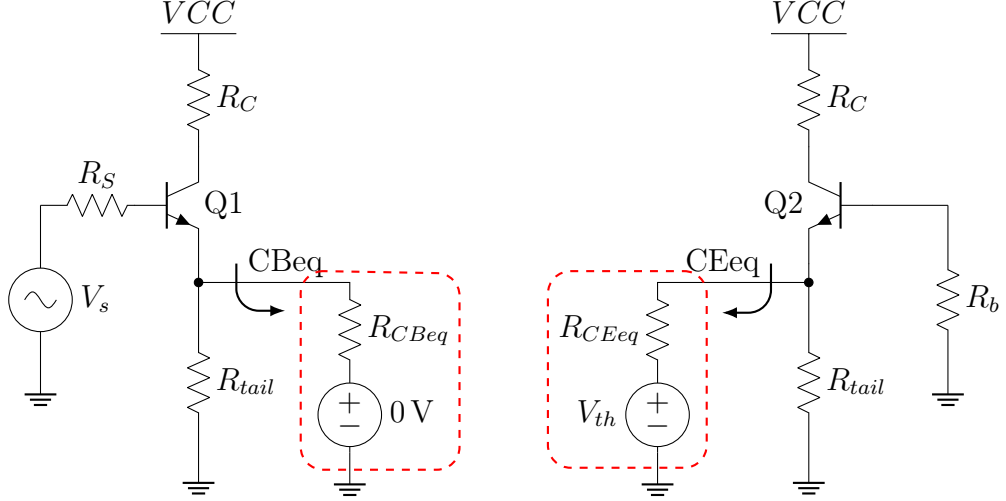


Figure 1: Schematic of the differential amplifier analysis with Thevenin equivalence

The Gains of the stages can be calculated by applying Thevenin's theorem to half part of the circuit at a time [?] [?]. Ignoring the contributions of R_{tail} , the CE amplifier sees from its emitter an equivalent source and resistance, representing the other half branch. The equivalent voltage source is zero because there is not an input signal applied at the base of Q2, meanwhile the equivalent resistance matches with the input resistance of the CB:

$$R_{CBeq} = \frac{r_\pi + R_b}{g_m r_\pi + 1} \quad (1)$$

(?) can also be written as $r_e \left(1 + \frac{R_b}{r_\pi}\right)$ [?], at the condition of $g_m r_\pi \gg 1$.

The resulting CE gain is defined as:

$$A_{CE0} = -\frac{\alpha R_C}{R_E + R_{CBeq}} = -\frac{\alpha R_C}{\left(r_e + \frac{R_s}{g_m r_\pi + 1}\right) + \frac{r_\pi + R_b}{g_m r_\pi + 1}} \quad (2)$$

The CB can be analyzed in the same way of the CE with the difference for which this time with $V_{th} \approx V_s$. The voltage gain can be calculated as the result of two blocks connected in cascade [?]: the Thevenin block, which contributes with a value $A_{path} = \frac{V_s}{V_e}$, and the remaining block consisting of the open circuit voltage

gain of the CB $A_{CBv0} = \frac{V_o}{V_e}$. The resulting gain for the CB stage is defined as:

$$A_{CB0} = A_{path} \cdot A_{CBv0} = \frac{\alpha R_C}{\left(r_e + \frac{R_b}{g_m r_\pi + 1}\right) + \frac{r_\pi + R_S}{g_m r_\pi + 1}} \quad (3)$$

(??) and (??) are equal in module and opposite in phase, effectively creating a balanced output signal. These results are only valid at low frequencies. in the D band, The the gain is calculated in the same but the instrinsic capacitors must be taken into account. The new impedances changes according to the topology of amplifier is taken into consideration. For the CE amplifier is know that:

$$Z_{\pi,CEeff} = r_\pi \parallel (sC_{be} + sC_{bc}(1 + |A_{CE0}|)) \quad (4)$$

$$Z_{E,CEeff} = r_e + \frac{R_S}{g_m Z_{\pi,CEeff} + 1} \quad (5)$$

$$Z_{C,CEeff} = Z_C \parallel sC_{bc} \left(1 + \frac{1}{|A_{CE0}|}\right) \quad (6)$$

the same results are applied for the CB amplifier but without the miller

$$Z_{\pi,CBeff} = r_\pi \parallel sC_{be} \quad (7)$$

$$Z_{E,CBeff} = r_e + \frac{R_b}{g_m Z_{\pi,CBeff} + 1} \quad (8)$$

$$Z_{C,CBeff} = Z_C \parallel sC_{bc} \quad (9)$$

(??) and (??) can be updated considering that

$$Z_{CBeq} = \frac{Z_{\pi,CBeff} + R_b}{g_m Z_{\pi,CBeff} + 1} \quad Z_{CEeq} = \frac{Z_{\pi,CEeff} + R_b}{g_m Z_{\pi,CEeff} + 1} \quad (10)$$

and so the gain is:

$$A_{CE}(\omega) = - \frac{\alpha Z_{C,CEeff}}{\left(r_e + \frac{R_S}{g_m Z_{\pi,CEeff} + 1}\right) + \frac{Z_{\pi,CEeff} + R_b}{g_m Z_{\pi,CEeff} + 1}} \quad (11)$$

$$A_{CB}(\omega) = \frac{\alpha Z_{C,CBeff}}{\left(r_e + \frac{R_b}{g_m Z_{\pi,CBeff} + 1}\right) + \frac{Z_{\pi,CBeff} + R_S}{g_m Z_{\pi,CBeff} + 1}} \quad (12)$$

The voltage gain of a network can also be expressed as a function of Z , Y , or other parameters. In particular, the voltage gain of a two-port network [?], terminated with a load admittance (commonly $Y_L = 0.02\text{ S}$), is given by

$$A_v = \frac{-Y_{21}}{Y_{22} + Y_L} \quad (13)$$

For a multi-port network, and in particular for a three-port, (??) can be extended by simultaneously loading the output ports such that $I_2 = -Y_L V_2$ and $I_3 = -Y_L V_3$. This leads to the complete expression for the amplitude imbalance, expressed in linear scale, of a single-input balanced-output differential amplifier:

$$\frac{A_{vCE}}{A_{vCB}} = -\frac{Y_{21}}{Y_{31}} \cdot \frac{Y_{33} + Y_L - Y_{31}Y_{23}}{Y_{22} + Y_L - Y_{21}Y_{32}} \quad (14)$$

In a first-order approximation, by neglecting the contributions $Y_{31}Y_{23}$ and $Y_{21}Y_{32}$, the amplitude imbalance can be rewritten as in (??) where, in short, it depends on the product of two ratios: the ratio of the forward gains and the ratio of the output load terms.

$$\left| \frac{A_{vCE}}{A_{vCB}} = \frac{Y_{21}}{Y_{31}} \cdot \frac{Y_{33} + Y_L}{Y_{22} + Y_L} \right| \quad (15)$$

The simplest forms of the Y -parameters are those of the intrinsic CE and CB small-signal π -models [?]. With further simplifications, the following expressions are obtained:

$$Y_{CE} = \begin{pmatrix} Y_\pi + j\omega C_{bc} & -j\omega C_{bc} \\ g_m - j\omega C_{bc} & g_0 + j\omega C_{bc} \end{pmatrix} \quad Y_{CB} = \begin{pmatrix} g_m + g_0 + Y_\pi & -g_0 \\ -g_m - g_0 & g_0 + j\omega C_{bc} \end{pmatrix} \quad (16)$$

For the purposes of this work, only the CB case is considered, and the following analysis provides an approximate description of the behaviour of its parameters. The intrinsic small-signal π -model can be extended to an extrinsic model by including an external base admittance Y_B . The resulting elements of the Y -parameter matrix are denoted by $Y_{ji,\text{eff}}$. The equations for the CB configuration are reported in Table ??, together with the small-signal circuits used for their derivation, from which further conclusions can be drawn. For large values of Y_B , such as when a large capacitor is used, $Y_{11,\text{eff}}$ and $Y_{22,\text{eff}}$ tend toward their maximum values, $g_0 + g_m + Y_\pi$ and $g_0 + Y_\mu$, respectively. These values coincide with those of the ideal CB amplifier reported in (??). On the other hand, for lower values of $j\omega C_B$, Y_B decreases, and the second term in both $Y_{11,\text{eff}}$ and $Y_{22,\text{eff}}$ decreases as well.

If Y_B is inductive instead, the behaviour is reversed.

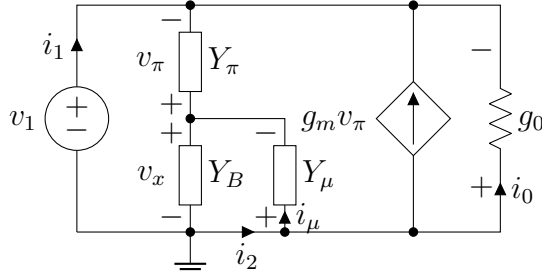
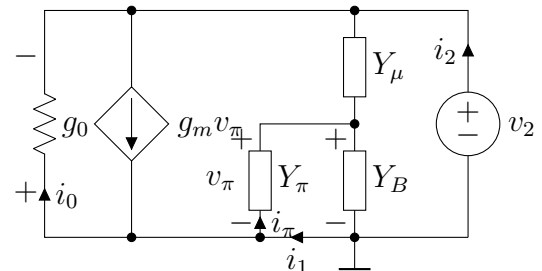
	
$Y_{11,\text{eff}} = g_0 + (g_m + Y_\pi) \frac{Y_B + Y_\mu}{Y_B + Y_\pi + Y_\mu} \quad (17)$	$Y_{21,\text{eff}} = -g_0 - \frac{g_m(Y_B + Y_\mu) + Y_\pi Y_\mu}{Y_B + Y_\pi + Y_\mu} \quad (18)$
	
$Y_{12,\text{eff}} = -g_0 - \frac{g_m Y_\mu + Y_\pi Y_\mu}{Y_B + Y_\pi + Y_\mu} \quad (19)$	$Y_{22,\text{eff}} = g_0 + \frac{Y_\mu(Y_B + Y_\pi + g_m)}{Y_B + Y_\pi + Y_\mu} \quad (20)$

Table 1: Extrinsic Y parameters derived from the intrinsic BJT small-signal model with the addition of Y_B at the base.

1.2 Cascode current mirror

(??), (??), (??) and (??) are made in the condition of $R_{\text{CBeq}}(Z_{\text{CBeq}})$, $R_{\text{CEeq}}(Z_{\text{CEeq}}) \ll R_{\text{tail}}(Z_{\text{tail}})$, permitting the remotion of the R_{tail} influence from

the equations. The easiest way to realize such condition is to implement a very big resistance value but such choice is never implemented as it increases the dissipated power, reduces the voltage headroom and introduces parasitics in compact designs. High impedance at the emitter is commonly achieved with a current mirror, among which a cascode performs even better. Furthermore, to compensate the decline of the output impedance in D band, a shunt capacitor at the base of the Q3 transistor (Figure ??) can be applied.

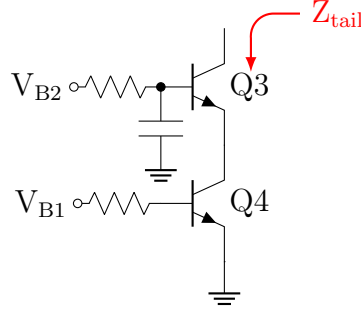


Figure 2: Schematic of the cascode current mirror without the left side biasing network

An explanation of this choice can be given by analyzing the output impedance of a CE and a CB topology. At higher frequencies, the discrepancy between the CE and CB output impedance becomes greater because of the different weight of the intrinsic collector base capacitance defined C_{CB} . In fact, If the amplifier works as a CE, the collector-base capacitance can be split with the miller theorem and the resultant output will be

$$Z_{out,CE} \approx r_o \parallel \frac{1}{j\omega C_{CB} \left(1 + \frac{1}{|A_{v,CE}|}\right)} \quad (21)$$

The Common Base amplifier has the base ideally fixed at ground ($Z_s \approx \frac{1}{sC}$) and hence, the capacitance can be taken out to the collector as a shunt element. In this situation the output impedance is given by a parallel between the impedance and the intrinsic collector-base capacitor.

$$Z_{out,CB} \approx r_o \parallel \frac{1}{j\omega C_{CB}} \quad (22)$$

As a result, the Z_{out} is slightly greater at frequencies above the input pole compared to the CE amplifier. More importantly, the CB configuration does not enhance the collector–base capacitor. To add further reasons, the implementation of the shunt capacitor permits to recreate a $\frac{1}{g_m}$ virtual load for Q1, thus creating an ideal unity gain and reducing the miller capacitance effect of this component.

1.3 Base inductor degeneration

Taking the CB amplifier and replacing the shunt capacitor with an inductor is one of the topologies used to create an oscillator. In the context of this work, the oscillator theory is not going to be covered but, still, a study is necessary to understand how suitable values of L can be used to tune the impedances while maintaining the circuit of Figure ?? in the amplifier configuration.

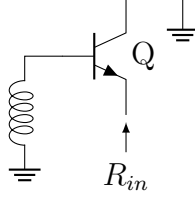


Figure 3: Schematic of the impedance tuning method using a base inductance

The input impedance can be extracted from $Y_{11,eff}$ and rewritten into the form present in [?], chapter 10. For this purpose all the simplifying assumptions indicated by the author will be maintained, starting from ignoring the emitter impedance and the output conductance g_0 :

$$Y_{11,eff} = \left(\frac{g_m}{Y_\pi} + 1 \right) \frac{Y_\pi(Y_B + Y_\mu)}{Y_B + Y_\pi + Y_\mu} = \left(\frac{g_m}{Y_\pi} + 1 \right) \frac{1}{\frac{Y_B + Y_\pi + Y_\mu}{Y_\pi(Y_B + Y_\mu)}} \quad (23)$$

$$= \left(\frac{g_m}{Y_\pi} + 1 \right) \frac{1}{\frac{1}{Y_\pi} + \frac{1}{Y_B + Y_\mu}} \quad (24)$$

$$Z_{in,eff} = \frac{1}{Y_{1,eff}} = \frac{\frac{1}{Y_\pi} + \frac{1}{Y_B + Y_\mu}}{\left(\frac{g_m}{Y_\pi} + 1 \right)} \quad (25)$$

By calling $Z_\pi = \frac{1}{Y_\pi}$ and $Z_{Bx} = \frac{1}{Y_B + Y_\mu}$ the equation in the book [?] is obtained in (??)

$$Z_{in,eff} = \frac{Z_{Bx} + Z_\pi}{1 + g_m Z_\pi} = \frac{(1 + j\omega C_\pi Z_{Bx})(g_m + j\omega C_\pi)}{g_m^2 + \omega^2 C_\pi^2} \quad (26)$$

Following the author in [?], by assuming $(f_T/f)^2 \gg 1$, so to say $g_m^2 \gg \omega^2 C_\pi^2$, the

real part of the impedance can be further simplified.

$$\mathbf{Re}(Z_{in,eff}) \approx \frac{1}{g_m} - \frac{\omega^2 C_\pi L_b}{g_m(1 - \omega^2 C_{bc} L_b)} \quad (27)$$

The resistance (??) reduces with respect to L, becoming negative for

$$L > \frac{1}{\omega^2(C_\pi + C_{bc})} \quad (28)$$

The evaluation of the output impedance can be achieved by the same means, with no citation provided.

$$Y_{22,eff} = \frac{Y_\mu(Y_B + Y_\pi + g_m)}{Y_B + Y_\pi + Y_\mu} = \frac{Y_\mu(Y_B + Y_\pi)}{Y_B + Y_\pi + Y_\mu} \left(1 + \frac{g_m}{Y_B + Y_\pi}\right) \quad (29)$$

$$= \frac{1}{\frac{1}{Y_\mu} + \frac{1}{Y_B + Y_\pi}} \left(1 + \frac{g_m}{Y_B + Y_\pi}\right) \quad (30)$$

then

$$Z_{out,eff} = \frac{1}{Y_{22,eff}} = \frac{\frac{1}{Y_\mu} + \frac{1}{Y_B + Y_\pi}}{1 + \frac{g_m}{Y_B + Y_\pi}} \quad (31)$$

By calling $Z_\mu = \frac{1}{Y_\mu}$ and $Z_{Bx} = \frac{1}{Y_B + Y_\pi}$,

$$Z_{out,eff} = \frac{Z_{bc} + Z_{Bx}}{1 + g_m Z_{Bx}} \quad (32)$$

if $g_m \gg 1/Z_{Bx}$ then:

$$Re(Z_{out,eff}) = \frac{1}{g_m} - \frac{1 - \omega^2 C_\pi L}{\omega^2 C_\mu L g_m} \quad (33)$$

From now on there is an error on the inequalities. the approximated resistance can increase or decrease depending by the second term. In particular, the resistor increases if the numerator is negative, hence if

$$L < \frac{1}{\omega^2 C_\pi} \quad (34)$$

For simplicity we could also consider a safeguard value by placing

$$L < \frac{1}{\omega^2(C_\pi + C_\mu)} < \frac{1}{\omega^2 C_\pi} \quad (35)$$

(??) and (??) guarantee the same conditions under which the real part of Z_{in} in (??) decreases while remaining positive, and under which the real part of Z_{out} (??) increases.

1.4 S-parameters

As the frequency increases, the wavelength of the electromagnetic waves reduces and becomes comparable to the circuits dimension. In this situation, the lumped theory approximation, which states that the signal is constant and measurable in every point of the space in a given moment of time, is no longer valid, introducing the concept according to which the Voltage and the current moves from one point to another as an electromagnetic wave constituted by the sum of incident and reflected terms. Given the dependency of the waves from the space and time in which the measurements are taken, and combining this with the fact that parasitic components introduced by the measuring instruments interfere with the DUT behaviour, the physical measurements of voltages and currents lead to incorrect values and are, therefore, avoided alongside the Z, Y, ABCD and other parameters. The Scattering parameters, S-parameters for short, are introduced for RF application purposes and are defined as the ratio between the reflected and the incident wave on one port when all the others are terminated to a Z_0 load, commonly 50Ω , to eliminate the contribution from different sources.

A two port network can be described as a system of two equations and can be represented in a matricial form:

$$\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \cdot \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \quad (36)$$

where

$$S_{ji} = \left. \frac{b_j}{a_i} \right|_{a_j=0} \quad (37)$$

The S parameters are best suited for describing the network at high frequencies because they require fixed loads and therefore are more reliable in comparison to parameters that depend on a perfect short or a perfect open circuit. Despite that, with mathematical manipulation, it is still possible to jump from S parameters to

a different one or vice versa.

In this work, unless otherwise stated, the amplitude imbalance and phase imbalance parameters are defined as follows:

$$\Delta\varepsilon(\text{dB}) = S_{21}(\text{dB}) - S_{31}(\text{dB}) \quad (38)$$

$$\Delta\phi(\text{deg}) = \text{unwrap}(\text{phase}(S_{21})) - \text{unwrap}(\text{phase}(S_{31})) \quad (39)$$

1.5 Stability

Through the use of the S parameters, it is possible to define the stability behaviour of the network taken into consideration [?][?]. A two-port network is defined as unconditionally stable if both the magnitudes of the input and output reflection coefficients are less than unity for every source and load termination over the operating frequency range. If this condition is not satisfied, the two-port network is defined as conditionally unstable, or equivalently conditionally stable, meaning that at least one value of the source or load termination causes oscillations and therefore must be avoided, unless the circuit is intended to operate as an oscillator.

The stability of a two-port network can be described by the stability factor K and Δ :

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2}{2|S_{12}S_{21}|} \quad (40)$$

$$\Delta = S_{11}S_{22} - S_{12}S_{21} \quad (41)$$

A two-port network is said to be unconditionally stable if $K > 1$ and $|\Delta| < 1$. Equivalently, using the Edwards-Sinsky stability factors, unconditional stability is obtained when $\mu > 1$ and $\mu' > 1$. If these conditions are not satisfied, the network is considered conditionally stable.

The Edwards-Sinsky criterion provides a necessary and sufficient condition for unconditional stability. In particular, both $\mu > 1$ and $\mu' > 1$ must be satisfied. μ evaluates the stability with respect to the source termination, while μ' evaluates the stability with respect to the load termination.

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12}S_{21}|} \quad (42)$$

$$\mu' = \frac{1 - |S_{22}|^2}{|S_{11} - \Delta S_{22}^*| + |S_{12}S_{21}|} \quad (43)$$

1.6 Gain

A network's power gain can be defined with three parameters: available power gain, Operating power gain and transducer power gain.

The available power gain is expressed as the ratio between the power available from the network, so the power delivered from the network to the load (P_{AVN}), and the power available from the source (P_{AVS}). In other words, the G_A is a parameter that depends from input matching network as well as the source impedance, while it is independent from the load.

The operating power gain G_P can be seen as the opposite of G_A and is defined as the ratio between the power delivered to the load (P_L) and the power input to the network (P_{in}), which means that it is independent from the source side and is dependent from the load matching network and termination.

The transducer gain is defined as P_L/P_{in} and therefore represents, among the three, the most complete parameter as it considers both source and load impedances [?] and so it describes the entire power transfer from input to output of a one-stage device. In simultaneous conjugate matching conditions, G_T , G_A , and G_P coincide.

1.6.1 Stability gain dependence for simultaneous conjugated networks

The general input reflection coefficients from the source and the load termination [?][?] are

$$\Gamma_{in} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L} \quad (44)$$

$$\Gamma_{out} = S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1 - S_{22}\Gamma_S} \quad (45)$$

When the two-port network is unilateral, so for $S_{12} = 0$, the reflection parameters simplify, becoming $\Gamma_{in} = S_{11}$ and $\Gamma_{out} = S_{22}$.

In this condition G_T takes the name of unilateral transducer power gain G_{UT} . Its maximum value, in (??), is achieved by applying simultaneous conjugate matching at the input and output, so as to maximize G_s and G_L .

$$G_{UTmax} = \underbrace{\frac{1}{1 - |S_{11}|^2}}_{G_{Smax}} \cdot \underbrace{|S_{21}|^2}_{G_0} \cdot \underbrace{\frac{1}{1 - |S_{22}|^2}}_{G_{Lmax}} \quad (46)$$

In a realistic case study, a two-port network is not perfectly unilateral, but it can be treated as such if S_{12} is sufficiently small. Furthermore, the dependence of G_T on the stability factors must be taken into account.

In general, the maximum power gain is achieved when the simultaneous matching network is applied at both the source (with $\Gamma_S = \Gamma_{in}^*$) and load ($\Gamma_L = \Gamma_{out}^*$). the reflection coefficients which permit this match can be called Γ_{MS} and Γ_{ML} and, from the stability point of view, the two port network is unconditionally stable if $|\Gamma_{MS}|$ and $|\Gamma_{ML}|$ are less than 1, which can be traced back to the condition $K > 1$ and $|\Delta| < 1$.

The Keysight ADS parameter "*MaxGain*" [?] refers to the maximum gain associated with a two-port network and, based on classical microwave amplifier theory[?][?], it corresponds to either the maximum available gain or the maximum stable gain, depending on the stability condition of the device. In particular, when the network is unconditionally stable, MaxGain corresponds to the MAG.

$$MAG = G_{T,max}|_{K>1} = \frac{|S_{21}|}{|S_{12}|}(K - \sqrt{K^2 - 1}) \quad (47)$$

When the network is conditionally stable, MaxGain corresponds to the MSG.

$$MSG = G_{T,max}|_{K=1} = \frac{|S_{21}|}{|S_{12}|} \quad (48)$$

From (??) and (??), it follows that the MSG is the limiting value of the MAG for $K = 1$ and, for greater values of K , the MAG decreases.

1.7 Mixed mode parameters

The S parameters can be edited to achieve a profound analysis of differential networks. To do so, the power waves, the normalized voltages and the normalized currents of each port are rewritten by using differential mode and common mode components [?]. This permits to change the S matrix in a different form, taking the name of mixed mode S parameters, abbreviated as S_{mm} .

For this work, the mixed mode parameters are introduced mainly as a tool to extrapolate the CMRR of the devices, by mean of MATLAB simulation tool, S parameter (S_{ds}) and single input to common mode output S parameter (S_{cs}).

$$\mathbf{S}_{mm} = \begin{bmatrix} S_{dd} & S_{dc} \\ S_{cd} & S_{cc} \end{bmatrix} \iff \mathbf{S}_{se} = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \quad (49)$$

The conversion starts from a general view. The objective is to convert a three-port network, such as a balun, into a new form containing differential and common modes parameters. The conversion is performed through an algebraic transforma-

tion, obtained by multiplying the single-ended S-parameter matrix S_{se} by M and M^{-1} , where M and M^{-1} are the conversion matrices.

$$\mathbf{S}_{mm} = \mathbf{M} \mathbf{S}_{se} \mathbf{M}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad (50)$$

Resulting in the matrix:

$$\begin{bmatrix} S_{11} & \frac{1}{\sqrt{2}}(S_{12} - S_{13}) & \frac{1}{\sqrt{2}}(S_{12} + S_{13}) \\ \frac{1}{\sqrt{2}}(S_{21} - S_{31}) & \frac{1}{2}(S_{22} - S_{32} - S_{23} + S_{33}) & \frac{1}{2}(S_{22} - S_{32} + S_{23} - S_{33}) \\ \frac{1}{\sqrt{2}}(S_{12} + S_{13}) & \frac{1}{2}(S_{22} + S_{32} - S_{23} - S_{33}) & \frac{1}{2}(S_{22} + S_{32} + S_{23} + S_{33}) \end{bmatrix} \quad (51)$$

which can be called as follows

$$\mathbf{S}_{mm} = \begin{bmatrix} S_{ss} & S_{sd} & S_{sc} \\ S_{ds} & S_{dd} & S_{dc} \\ S_{cs} & S_{cd} & S_{cc} \end{bmatrix} \quad (52)$$

where the most important parameters needed to CMRR are S_{ds} and S_{cs}

$$\text{CMRR}(dB) = 20 \log_{10} \frac{|S_{ds}|}{|S_{cs}|} \quad (53)$$

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